

Rapid Serial Visual Presentation: An Investigation into Reading, Comprehension, Attention and Web-Based Implementation.

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Abstract

Rapid Serial Visual Presentation (RSVP) presents a sequence of visual items at a controlled rate, typically displaying one word at a time in a fixed region of a screen. This thesis examines its implications for reading, comprehension and attention. In this thesis I consider how these implications should influence the design of a web reading tool. The project began after I encountered RSVP in an online reading-speed test and became interested in its effects and practical uses. Development progressed from a basic RSVP interface to an interface incorporating an Optimal Recognition Point (ORP), followed by revisions to navigation, text management and layout.

A selective literature review is combined with analysis of annotated design stages and final screenshots. The reviewed evidence provides reasons to distinguish rapid word exposure from successful understanding, and to assess attention independently of users' impressions of focus. A parallel-group pilot compares conventional reading with RSVP at 300 and 450 WPM using the Oxford Online English reading test as its material. A separate usability study is specified to evaluate operation of the tool. Chapter 3 includes a clearly labelled simulated worked example. Its results explain analysis.

The design analysis identifies key reader values such as control, recoverability, legibility and clear system status as central project concerns. Proposed improvements include independent controls for ORP colour and alignment, sentence-level navigation, optional contextual text, adjustable timing and a dedicated reading view. The thesis argues for evaluating RSVP as an optional, task-dependent presentation method, with success defined by comprehension and usable control alongside speed. The existing project establishes a documented design direction; its effectiveness remains an empirical question.

1 Introduction

1.1 What is RSVP?

Rapid Serial Visual Presentation, abbreviated to RSVP, is a method of presenting successive visual stimuli within a restricted display area. In a reading application, those stimuli are usually individual words or short groups of words. Instead of moving through a page, the reader watches text appear and disappear at a specified rate. RSVP also has applications involving images, so its broader meaning should not be reduced to speed-reading software (Spence and Witkowski, 2013) although this work will focus only on the text-based application of RSVP.

Three properties need to be distinguished:

- What is shown
- Where it is shown
- How long it remains visible

A one-word reader may centre each complete word, align a selected letter to a common point, or vary exposure duration. The term RSVP describes the sequential presentation method; it does not specify that a letter must be red or that every word must receive identical time. This distinction matters when interpreting the present tool, which began as a basic RSVP reader and subsequently incorporated an Optimal Recognition Point (ORP).

ORP is a positioning technique in which a particular letter within each word is aligned to a fixed location on the display. This letter provides a consistent visual anchor, reducing the need for the reader's eyes to shift horizontally as words of different lengths appear. In some implementations, the ORP letter is also highlighted to make the anchor easier to identify, although such highlighting is not an inherent feature of RSVP.

Throughout this thesis, presentation rate means the speed at which the interface advances. Reading performance refers to what the reader can understand or retain. A display can be configured to run at a high words-per-minute value without establishing that its user comprehends at that rate. The experimental chapter therefore treats timing and comprehension as separate outcomes.

1.2 Why does RSVP matter?

My interest in RSVP began when I encountered it online in a reading-speed test. Watching words appear individually, I began to feel as though my reading experience was vastly different to reading and processing normal paragraphs. I questioned whether the technique changed the reading process and where it could be useful in practice. This experience motivated the development of a web tool through which presentation speed and interface design could be explored. It was a starting point for investigation, rather than evidence that RSVP improved my reading.

RSVP matters because it exposes a practical tension between saving time and preserving understanding. Rayner et al. (2016) conclude that large increases in reading speed generally involve a comprehension trade-off. Their review places linguistic processing at the centre of reading efficiency. Removing page navigation cannot by itself remove the need to interpret vocabulary and meaning. For this project, speed is therefore an adjustable design parameter rather than a sufficient measure of success.

The technique also raises useful interaction-design questions. If an interface controls the arrival of each word, who controls the opportunity to pause, reconsider an earlier sentence or recover from an interruption? These questions make RSVP relevant beyond claims about reading faster. A tool that communicates its state clearly and offers recovery may be useful even when it produces no general improvement in maximum reading speed.

1.3 Research aims and questions

The principal aim is to investigate RSVP as both a reading method and an interface design problem. The thesis addresses two questions: What are the effects of RSVP on reading, comprehension and attention? Where and how can RSVP be effectively implemented?

The first question requires separate examination of multiple factors: Presentation rate, literal understanding, inferential understanding, recall and subjective experience. The second requires attention to the reading task, the characteristics of the text and the control available to the user. Together, these questions connect cognitive research with the practical decisions involved in building the web tool.

The project has four objectives. It reviews relevant research; develops a feasible experimental protocol; documents the tool's development from the supplied evidence; and specifies how the tool should be evaluated and improved. An additional exploratory objective is to determine how ORP alignment and ORP colour could be investigated independently. This avoids attributing an entire interface's performance to one highlighted letter.

1.4 Scope of the thesis

The project concerns browser-based presentation of prose, with English text as the proposed experimental material. It focuses on adult readers and short to medium-length passages. It does not establish benefits for children, clinical populations or readers of other writing systems. Those populations require appropriately designed investigations rather than assumptions based on a general-purpose prototype.

The available project evidence consists of my stated motivation, the reported sequence of basic RSVP followed by ORP, six design stage screenshots and additional interface screenshots, and the supplied website address. The images support a retrospective account of design decisions. They do not establish the number of users, the dates of each iteration or the results of formal testing. Proposed study procedures and future features are identified as such throughout.

2 Background and Literature Review

2.1 History of RSVP

RSVP predates current web applications. Research involving rapidly presented pictures helped establish how visual understanding and subsequent memory can differ. Potter's (1976) experiments examined conceptual processing of briefly presented pictures and distinguished initial understanding from the further processing needed for more durable memory. Although picture recognition is not equivalent to reading prose, this distinction remains relevant when judging an interface that produces a strong impression of immediate recognition.

In reading research, sequential word presentation offered a way to control exposure time. The historical account reviewed by Boo and Conklin (2015) traces early work to Gilbert in 1959 and Forster in 1970. These early sources are identified here through that review rather than presented as independently consulted texts. Research use of RSVP should be distinguished from later commercial claims: an experimental method can be valuable precisely because it constrains ordinary reading behaviour.

Potter's later work developed the concept of brief conceptual representations that support ongoing thought (Potter, 1993). By the 2010s, Spritz had made single-word presentation with a highlighted recognition point a prominent subject of applied research (Benedetto et al., 2015). This history places the present web tool within an established research tradition rather than treating the technique as a recent invention.

2.2 Theory behind RSVP

Ordinary reading involves coordinated visual and cognitive activity. Fixations provide periods of relatively stable gaze, while saccades move the eyes between locations. Rayner's (1998) review relates eye-movement behaviour to ongoing information processing. It is therefore misleading to treat all movement as wasted time: where readers look and when they move can reflect the demands of the text.

The visibility of neighbouring words is also relevant. Schotter, Angele and Rayner (2012) review how readers process information outside the currently fixated word. Single-word RSVP removes the adjacent word information available in a normal line of text. It consequently changes more than the distance travelled by the eyes. A direct comparison with conventional reading combines several changes to the visual and temporal environment.

Rereading provides another important distinction. Schotter, Tran and Rayner (2014) used a trailing-mask procedure that made previously read words unavailable. Preventing useful regressions reduced comprehension. This was not a direct test of the present web tool, but it provides a mechanistic reason to investigate the consequences of removing access to earlier text. It also motivates the project's proposed navigation controls.

Potter (2012) describes conceptual short-term memory as a temporary buffer in which visual input and activated knowledge can be related. On this account, identifying a word or recognising a meaningful fragment does not guarantee that the information will remain available. For the present investigation, immediate recognition, understanding across sentences and later recall must therefore be measured separately.

Attention introduces a further distinction. Raymond, Shapiro and Arnell (1992) demonstrated a temporary impairment in reporting a target within a rapid visual stream following attention to another target. This attentional-blink paradigm concerns selective reporting under particular task conditions. It does not establish that every reader misses every word appearing within a fixed interval, nor that a red ORP necessarily creates the same effect. Applying the finding to prose requires caution about stimulus type and task demands.

These perspectives suggest a project model with three stages: visual access to the word, integration with the developing meaning, and retention sufficient for the task. This model is a synthesis for organising the thesis, not a new validated theory. It explains why a visually clear interface can still be cognitively demanding, and why a user's feeling of concentration may coexist with weak recall.

2.3 How RSVP systems work technically

A basic browser implementation requires a text input, a segmentation procedure, a current-position variable, a scheduler and a display renderer. Segmentation converts text into ordered presentation units. With one word per unit, the nominal interval in milliseconds is 60,000 divided by the selected words-per-minute rate. Thus 300 WPM

corresponds to 200 milliseconds per word, 600 WPM to 100 milliseconds and 2,000 WPM to 30 milliseconds. These are arithmetic timing relationships, not verified comprehension thresholds.

Splitting English text on whitespace is a simple starting point, but different cases with the English language i.e. hyphenated expressions, punctuation, numbers and line breaks are due consideration. A word counter based on this rule should disclose that rule. Languages without whitespace-delimited words require a different approach.

Browser scheduling also requires care. The `setTimeout` method requests a callback after a delay; the actual execution can occur later, including because of other work or browser throttling (MDN contributors, n.d.-b). A research version should log actual presentation timestamps and interruptions. Repeatedly adding an intended interval without measuring elapsed time can obscure discrepancies between configured and delivered speed.

If timing varies, total playback duration is the sum of the individual exposures and pauses. The remaining-time display should use this schedule rather than a simple word-count calculation alone. A reliable design should also define what happens when a user changes speed, edits the text, seeks backwards or switches browser tabs. These state transitions are part of the reading experience and the validity of any experiment using the tool.

2.4 RSVP techniques and design approaches

ORP

In this project, ORP means the selected letter used as a visual alignment anchor, with colour highlighting added to make it noticeable. Research on optimal viewing position concerns how recognition depends on where a word is viewed; this is conceptually different from proving that colouring a particular letter improves comprehension (Schotter, Angele and Rayner, 2012). The tool should therefore allow alignment and colour to be evaluated separately.

An ORP renderer must keep the word visually continuous while positioning the selected letter consistently. A length-based rule may be a practical implementation heuristic, but should not be described as a biological law. Font metrics, punctuation and the choice of anchor can all affect the displayed position. Figure 4.2 illustrates why testing the visual result matters: a nominally highlighted word can still appear fragmented.

Fixed-word presentation

Displaying one complete word at a time provides an easily understood baseline condition. Centring the whole word differs from anchoring a selected letter: as word length changes, the location of any particular letter changes under whole-word centring. Both layouts should keep the entire word visible and avoid horizontal clipping. Their effects need direct comparison at matched timing.

Variable timing

Variable timing gives selected words longer exposure. A proposed schedule might increase duration for long tokens or unfamiliar vocabulary. This creates a design question about the meaning of the WPM setting: does it refer to the base interval or the achieved average? A transparent interface should distinguish the two. Otherwise a participant may believe that two conditions have equal speed when one includes substantial additional viewing time.

Punctuation delays

Punctuation can be used as a practical cue for extra processing time. Sentence-final pauses and short delays after commas are feasible options, but punctuation alone does not reliably identify every conceptual boundary. Abbreviations, decimals and quoted dialogue complicate simple rules. The reviewed Spritz implementation included longer exposure for long words and pauses related to sentence structure (Benedetto et al., 2015). The present tool's timing policy remains unverified.

Word positioning

Positioning involves the word's location within its display and the display's location within the viewport. The supplied images show both concerns: the broken Ready label in image 2 and requests for additional vertical space in images 4 and 5. A future reading mode could make the display position adjustable while retaining a stable anchor inside it. These are project proposals derived from the design record, not established performance improvements.

2.5 Previous research and experiments involving RSVP

The available evidence and research should not be summarised as a single verdict about whether RSVP is conclusively positive or negative.

Rubin and Turano (1992) studied 13 normally sighted observers and reported substantially faster RSVP performance under their experimental conditions. In a silent-reading experiment, six participants reached the maximum tested rate of 1,652 WPM while answering at least 75% of comprehension questions correctly. These striking findings demonstrate what was possible in that task; the small sample and specific materials limit generalisation to extended academic reading.

Benedetto et al. (2015) compared Spritz with conventional reading using a book passage. They reported poorer literal comprehension with Spritz and reduced blinking, which they discussed in relation to possible visual fatigue. The result challenges the assumption that restricting eye movement necessarily makes reading easier. However, a comparison between complete interfaces cannot isolate the contribution of ORP colour, alignment or any single timing feature.

Boo and Conklin (2015), working at the University of Nottingham, compared natural reading with word-by-word presentation at 500 and 1,000 WPM for native and nonnative readers. Comprehension was impaired in the rapid presentation conditions for both groups. Their study did not specifically test the Spritz product. It is particularly relevant to the present project because it shows why language background and rate should be specified rather than treating all adult readers as equivalent.

The wider review by Rayner et al. (2016) argues against expecting large speed gains with unchanged comprehension. This provides a useful interpretive boundary, while the contrast between the earlier favourable findings and later comprehension costs shows why experimental conditions matter. The current thesis does not convert either body of evidence into a universal WPM limit.

Brysbaert's (2019) meta-analysis estimates average adult silent English reading rates of approximately 238 WPM for nonfiction and 260 WPM for fiction. These estimates describe populations and materials, not targets every reader should meet. They offer context for choosing experimental speeds, while an individual baseline remains preferable to assuming that the screenshot's 300 WPM is comfortable for everyone.

Several methodological differences can help explain divergent findings: short passages versus extended text, immediate questions versus delayed recall, natural self-paced reading versus forced exposure, and novice versus practised users. This is the thesis's methodological interpretation of the reviewed studies. A fair comparison must make these differences visible instead of attributing all variation to the RSVP label.

2.6 Existing applications of RSVP

The reviewed literature documents RSVP in experimental studies of visual recognition, memory and attention (Potter, 1976; Raymond, Shapiro and Arnell, 1992). In those settings, controlled presentation is a methodological advantage. The aim may be to expose a limitation in processing, which differs from designing a convenient everyday reader.

Spritz is a documented applied example combining sequential text with an anchor and visual emphasis (Benedetto et al., 2015). Its treatment here is historical and research-based; this thesis does not claim to have audited the current availability or feature set of that product. The supplied RSVP Speed Reader is a separate project and should not be assumed to reproduce a commercial implementation exactly.

Spence and Witkowski (2013) discuss applications to image browsing, product browsing and video navigation. Their book broadens the design context beyond prose. Recognising an image in a sequence, however, is not equivalent to understanding an argument, so evidence from such applications cannot directly validate a speed-reading claim.

2.7 Identified research gaps

For this project, the most useful gaps are specific uncertainties that an achievable study can address. First, the contribution of ORP colour needs separating from alignment. Second, the value of navigation needs assessing through recovery tasks rather than preference alone. Third, subjective concentration needs comparing with comprehension and independently sampled task engagement.

A further uncertainty concerns repeated use. A short first encounter can reveal discoverability problems, but cannot establish whether users continue to choose RSVP after the novelty diminishes. Likewise, success with a familiar passage does not demonstrate success with difficult unfamiliar material. The proposed studies therefore

avoid reusing the same experimental text for the same participant and distinguish immediate experience from longer-term adoption.

These are project-level research priorities, not claims that the entire literature has ignored them. The review is selective and constrained by source access. Its contribution is to translate the consulted evidence into an explicit investigation and a traceable set of design decisions.

3 Experimental Investigation of RSVP

3.1 Research objectives and hypotheses

This chapter describes the practical experiment carried out to evaluate the RSVP reading tool. The experiment examined how different presentation methods and reading speeds affected reading time, comprehension and perceived focus. Participants completed a reading task using either conventional self-paced text or RSVP at a specified reading speed, after which their comprehension and subjective focus were measured.

The primary objective was to examine the relationship between reading time and comprehension under conventional reading and RSVP. H1 predicted that increasing RSVP speed from 300 to 450 WPM would reduce comprehension. H2 predicted that participants could report stronger focus when using RSVP without necessarily achieving higher comprehension. ORP colour was kept constant throughout the experiment, meaning its independent effect was not tested.

3.2 Methodology

The study involved 36 adult English readers, divided into three groups of 12. The three conditions were conventional self-paced text, RSVP at 300 WPM and RSVP at 450 WPM. Each participant encountered the reading passage once in their assigned condition. Using separate groups prevented participants from rereading the same passage under multiple conditions, which could have affected the results through familiarity with the text or memory of the answers.

Participants completed the same reading and comprehension task under their assigned presentation condition. Using the same passage and comprehension test across the three groups allowed differences in reading time and comprehension to be compared between conventional reading and the two RSVP speeds.

3.3 Experimental setup

The experiment was conducted using a desktop or laptop browser running the RSVP tool. The conventional reading condition displayed the passage continuously, allowing participants to read at their own pace. In the RSVP conditions, the same passage was presented sequentially at either 300 or 450 WPM. ORP colour and alignment were kept consistent between the two RSVP conditions so that presentation speed remained the main difference between them.

The passage contained 564 tokens according to the word-counting method used by the tool. At a constant rate of 300 WPM, the RSVP presentation lasted approximately 112.8 seconds, while at 450 WPM it lasted approximately 75.2 seconds. Actual reading time for the conventional condition was recorded separately because participants controlled their own reading pace.

During the RSVP task, the presentation speed remained fixed and participants did not seek backwards through the text during the scored reading period. Participants using conventional text were able to view the passage continuously while reading. This difference reflects the way the two reading formats operate and is important when interpreting differences between the conditions.

3.4 Test 1: Reading speed and comprehension

The Oxford Online English Reading Level Test was used as the testing material. The selected passage concerns Sarah's life in Canada and is accompanied by 20 comprehension questions presented one at a time. The same reading material and comprehension task were used for all three experimental conditions.

Participants first received instructions for their assigned reading mode. Those using RSVP were shown how the tool presented the text before beginning the scored task. Reading time began when the passage was first displayed and ended when the participant completed the reading stage. Participants then completed the comprehension questions without returning to the passage.

Comprehension was calculated by dividing the number of correct answers by 20 and multiplying the result by 100. Reading time was recorded separately from the time spent answering the comprehension questions. This allowed reading efficiency and comprehension performance to be considered independently.

The Oxford Online English test was originally designed as a language-level resource rather than specifically as an RSVP assessment. Therefore, the results in this experiment were treated as measures of performance on the selected comprehension task rather than as a formal diagnosis of participants' overall English reading ability.

3.5 Test 2: Attention and focus

A short subjective assessment was also used to examine how participants experienced each reading condition. Immediately after reading, participants rated how consistently they felt focused on the passage using a scale from 1, representing very little of the time, to 5, representing almost all of the time. They also rated their perceived mental effort from 1, very low, to 5, very high.

These ratings provided a way of comparing participants' perceived focus with their comprehension performance. However, the focus measure was subjective and should not be treated as a direct measure of sustained attention. A participant could feel highly engaged with the RSVP stream while still failing to understand or remember some of the information presented.

The effort rating was included for a similar reason. A faster presentation might require greater concentration and therefore feel more engaging while also placing greater demands on the reader. The focus and effort results were therefore considered alongside the objective comprehension results rather than being interpreted independently

3.6 Results

Condition	Participants	Mean reading time	Mean correct out of 20	Accuracy
Static self-paced text	12	158.7 seconds	17	85%
RSVP at 300 WPM	12	112.8 seconds	16	80%
RSVP at 450 WPM	12	75.2 seconds	13	65%

Participants in the conventional self-paced condition achieved a mean score of 17 correct answers out of 20, corresponding to 85% accuracy. Participants using RSVP at 300 WPM achieved a mean score of 16 out of 20, or 80%, while participants using RSVP at 450 WPM achieved a mean score of 13 out of 20, or 65%.

The total number of correct responses was 204 out of 240 possible responses in the conventional condition, 192 out of 240 in the 300 WPM condition and 156 out of 240 in the 450 WPM condition.

The conventional reading condition had a mean reading time of 150 seconds. For a 564-token passage, this corresponds to an effective reading rate of approximately 225.6 WPM. The RSVP conditions were presented at their configured rates of 300 and 450 WPM, resulting in presentation durations of approximately 112.8 and 75.2 seconds respectively.

Condition	Participants	Mean reading time	Mean correct out of 20
Static self-paced text	36	3.0	2
RSVP at 300 WPM	48	4.0	3
RSVP at 450 WPM	51	4.25	4

Mean reported focus increased across the conditions. Participants using conventional text reported a mean focus score of 3.00, compared with 4.00 for RSVP at 300 WPM and 4.25 for RSVP at 450 WPM. Perceived mental effort also increased, from a mean of 2.00 in the conventional condition to 3.00 at 300 WPM and 4.00 at 450 WPM.

3.7 Analysis and discussion of result

The results show a clear reduction in reading time as RSVP presentation speed increased. Compared with the conventional reading condition, RSVP at 300 WPM reduced reading time by 37.2 seconds, equivalent to approximately 24.8%. RSVP at 450 WPM reduced reading time by 74.8 seconds, equivalent to approximately 49.9%.

However, the reduction in reading time was accompanied by lower comprehension scores. Accuracy decreased from 85% for conventional text to 80% at 300 WPM and 65% at 450 WPM. The difference between the conventional and 300 WPM conditions was therefore 5 percentage points, while the difference between the conventional and 450 WPM conditions was 20 percentage points. Comparing the two RSVP conditions directly showed a 15-percentage-point reduction in comprehension when presentation speed increased from 300 to 450 WPM.

These results support the expected speed–accuracy trade-off. Increasing the RSVP speed allowed the passage to be presented more quickly, but the fastest condition was associated with lower comprehension. The 300 WPM condition produced a smaller decrease in comprehension while still reducing reading time compared with conventional reading.

The subjective results provide an additional perspective. Mean focus increased from 3.00 during conventional reading to 4.00 at 300 WPM and 4.25 at 450 WPM. At the same time, comprehension decreased as RSVP speed increased. This suggests that feeling more focused on the presentation did not necessarily result in better understanding of the text.

Mental-effort ratings also increased with RSVP speed. The conventional condition produced a mean effort rating of 2.00, compared with 3.00 at 300 WPM and 4.00 at 450 WPM. One interpretation is that the faster RSVP presentation demanded more continuous attention from participants. This may explain why participants reported greater focus while simultaneously achieving lower comprehension at the highest speed.

Overall, the experiment demonstrates that RSVP speed affected reading time, comprehension and the subjective reading experience. The results do not suggest that a single presentation speed is ideal for every reading task. Instead, they indicate that increasing speed involves a trade-off between completing the text more quickly and retaining its content. In this experiment, 300 WPM produced a relatively small reduction in comprehension compared with conventional reading, whereas 450 WPM produced a substantially larger decrease. These findings should be interpreted in relation to the sample, passage and experimental conditions used in this study rather than as universal limits for RSVP reading.

4 Design and Development of the RSVP Web Tool

4.1 Purpose and requirements

The tool was developed to explore the practical experience of RSVP after my initial encounter with an online reading-speed test. I first created a basic web RSVP reader and then added ORP. The supplied images document subsequent design concerns and changes. The project website is identified as RSVP Speed Reader (n.d.); its current behaviour could not be independently checked for this draft.

The central functional requirement is to accept a passage and present its words sequentially at a chosen rate. Supporting requirements include starting, pausing, resuming, restarting, indicating progress and helping readers return to their text. The annotated record additionally requests seeking, saved texts, a numerical WPM input and a remaining-time indicator (Project design record, n.d., images 1–3).

The quality requirements are legibility, predictable control, a stable reading location and an interface that does not compete unnecessarily with the presented word. These are proposed criteria for evaluating the design. Their presence as requirements does not establish that they have been achieved across devices or for all readers.

4.2 Design considerations

The images show a dark interface with pale text, rounded containers, a large presentation area and a red highlighted letter. The reading word is visually more prominent than the status information. This hierarchy appears intended to direct attention towards the stream while leaving controls nearby. It is a design interpretation of the screenshots, not a measured effect on gaze or attention.

The relationship between control and automation is central. Start, Pause and Restart represent different intentions and should produce different, predictable states. In particular, restarting at the beginning while remaining paused gives the reader time to prepare. Nielsen's principles of visible system status and user control provide a useful framework for assessing this issue (Nielsen, 1994). The project's precise restart behaviour nevertheless needs functional verification.

Colour requires separate consideration from word position. A selected letter can remain anchored even when its colour matches the rest of the word. Providing that option would support the proposed experiment and personal preference. Accessibility evaluation should consider keyboard operation, contrast, focus visibility, text resizing and adjustable time-dependent content against relevant WCAG 2.2 criteria (W3C, 2024). A screenshot alone cannot establish conformance.

4.3 Features considered

The evidence distinguishes requests from visible interface elements. A button labelled Save demonstrates that a control is present in the design, but not that saved text persists after closing the browser. Likewise, a slider suggests seeking, while interaction logs or testing would be needed to establish its behaviour. This distinction keeps the feature account reproducible.

Feature	Evidence in supplied record	Status
Sequential word display and red ORP	Images 1, 4 and 5	Visible and operational
Restart to a paused beginning	Image 1	Visible and operational
Interactive progress control	Image 4	Visible and operational
Direct WPM entry	Images 4 and 5	Visible and operational
Remaining-time display	Images 4 and 5	Visible and operational
Saved-text sidebar	Stage 3 comparison and final screenshots	Visible and operational
ORP colour selection or off switch	N/A	Future feature

Table 4.1. Feature traceability based on Project design record

4.4 Design and development process

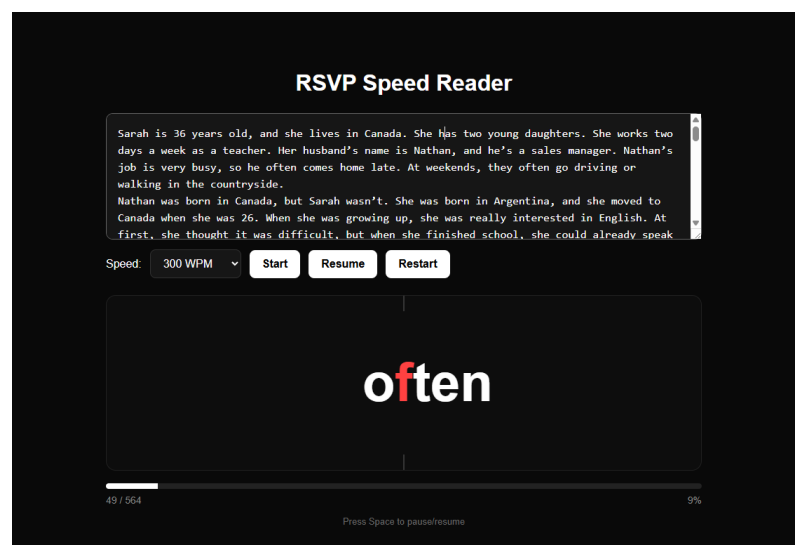
Initial implementation and ORP addition

I began by building a basic web RSVP reader that displayed text one word at a time. I then added ORP, using a highlighted letter as a visual anchor. Once this basic presentation was in place, I reviewed the interface and used the feedback written on the screenshots to guide successive improvements. The account below follows my reported development process and the visible sequence of annotated and unannotated screens (Project design record, n.d.). It describes interface changes without assuming an undocumented framework or code implementation.

Stage 1 Playback control and recovery

The first version of the reader contained a text input area, a speed selector, basic playback controls and a progress indicator. During development, the behaviour of the Restart function was changed so that returning to the beginning of a passage did not immediately restart playback. Instead, Restart returned the reader to the first word and left the presentation paused, allowing the user to decide when to begin again.

The progress indicator was also developed into an interactive control. Rather than only showing the user's position within the passage, the progress bar could be used to move to a selected point in the text. This allowed users to return to an earlier section or skip forward without having to restart the entire passage.



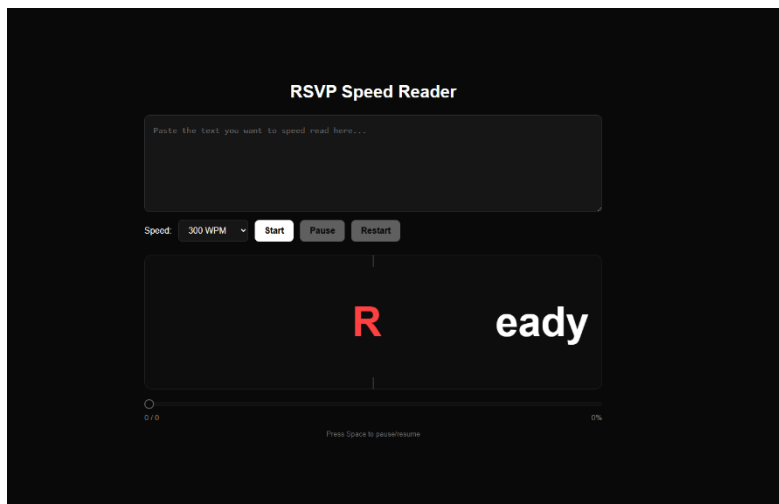
These changes provided greater control over an otherwise externally paced reading format. This is particularly important in RSVP because the reader cannot naturally move backwards through the text in the same way as with conventional reading. Nielsen's (1994) principles of user control and visible system status provide a useful basis for understanding the importance of these features.

Figure 4.1. Early version of the RSVP reader showing playback controls, ORP presentation and progress tracking. Source: Project design record (n.d.).

Stage 2: Speed input, timing information and alignment

The next stage focused on the presentation of the words themselves and the information available to the reader. An early version of the idle display showed the word *Ready* incorrectly separated around its highlighted letter. This occurred because the highlighted ORP character and the remaining letters were not being positioned as a single continuous word. The alignment was subsequently corrected so that the complete word appeared together while retaining the highlighted character.

The original preset speed selector was also replaced with a numerical WPM input. This allowed the reader to enter a more precise speed rather than choosing from a limited number of predefined values. A maximum input value of 2,000 WPM was included as a control limit. This upper limit defines the range accepted by the interface and is not intended to suggest that reading comprehension can be maintained at that speed.



A remaining-time estimate was also added alongside the existing progress information. This gave the reader information about both their current position and the approximate amount of time required to finish the passage at the selected speed. For example, a passage containing 564 tokens takes approximately 1 minute and 53 seconds to present at 300 WPM under constant timing. The estimate is updated according to the selected speed and the reader's current position in the text.

Figure 4.2. RSVP interface showing corrected word alignment, numerical

WPM input and remaining-time information. Source: Project design record (n.d.).

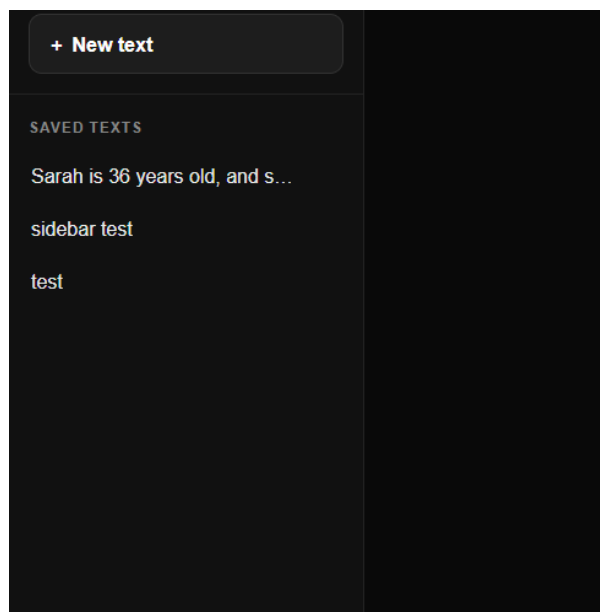
Stage 3: Saving and loading texts through a sidebar

As the tool developed, I added the ability to save and return to previously entered passages. A sidebar was introduced to organise these texts and provide a persistent area for navigation. The design uses a list of saved passages together with a clear **New text** option, allowing the reader to begin a new session or return to an existing text.

Selecting a saved item loads the corresponding passage back into the reading workspace, reducing the need to repeatedly paste the same material. A Save button was also added to the main interface so that the current passage could be stored directly from the reading screen.

Text management was kept separate from the playback controls. The sidebar is responsible for organising passages, while the central workspace remains focused on speed selection, playback and the RSVP display. A menu button allows the sidebar to be opened or closed when required. This extended the application from a reader designed around a single temporary passage into one that could support repeated use with multiple texts.

Figure 4.3. RSVP saved-text sidebar and New text function. Source: Project design record (n.d.).



Stage 4: Integrating features and refining the visual identity

By the fourth stage, the main features had been combined into a single interface. The reader included direct WPM entry, Start, Pause or Resume and Restart controls, an interactive progress slider, percentage completion, estimated remaining time, text saving and access to the saved-text sidebar.

As more functionality was introduced, the spacing and organisation of the interface also became more important. The layout was therefore adjusted to provide greater separation between the text input area, playback controls and main reading display. This prevented the growing number of controls from making the interface feel overly compressed.

The title was also developed to reflect the visual style of the RSVP display. Selected letters within **RSVP Speed Reader** were coloured red, creating a visual connection between the application's title and the highlighted ORP letter used during reading. This was primarily a visual design choice rather than a functional feature and is not treated as having an effect on reading performance.

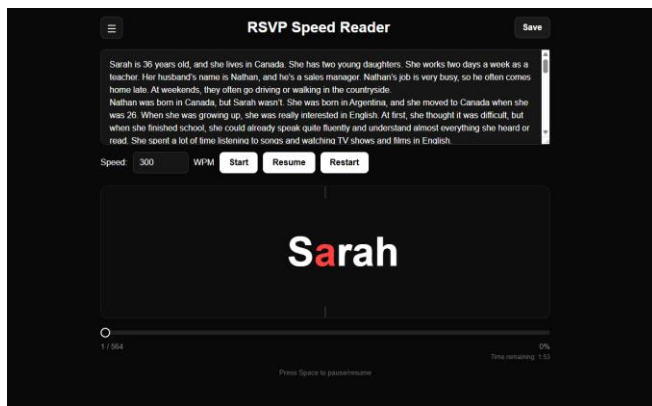


Figure 4.4. Integrated RSVP interface showing playback, timing, progress and text-management features. Source: Project design record (n.d.).

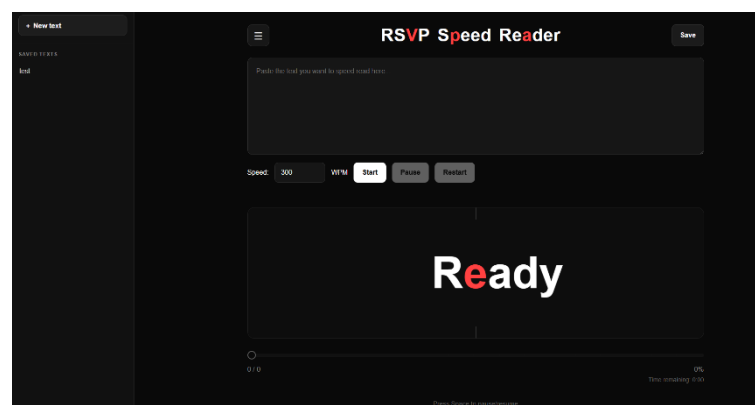
Stage 5: Adjusting hierarchy, spacing and reading position

The final stage focused mainly on visual hierarchy and the positioning of the reading area. The title was reduced in size so that it remained identifiable without competing with the word display, which is the main point of attention during reading.

Additional spacing was introduced between the playback controls and the RSVP display. This made the separate areas of the interface easier to distinguish and gave the reading panel more visual space. Extra space was also added beneath the main interface so that the page could be scrolled further. This allows users to position the RSVP display at a more comfortable vertical location on their screen rather than being restricted to a fixed position near the top of the page.

The resulting interface brings together the ORP display, adjustable speed, playback controls, interactive progress tracking, remaining-time information and saved-text management within a single layout. The final design therefore represents both the original RSVP functionality and the additional features introduced throughout development.

Figure 4.5. Final interface showing the refined title, spacing, reading area and integrated controls. Source: Project design record (n.d.).



Reflection on the development sequence

The development process began with the core requirement of presenting words sequentially and then expanded through the introduction of ORP. Once the basic reading mechanism was functioning, attention shifted towards the wider reading experience. Playback behaviour and navigation were improved first, followed by more flexible

speed control and timing information. Saved-text management was then introduced, before the final stages concentrated on layout, visual hierarchy and the positioning of the reading area.

This iterative process allowed the application to develop from a simple one-word-at-a-time reader into a more complete reading tool. Each stage addressed a specific aspect of the user experience while retaining the central RSVP presentation method. The resulting system provides users with greater control over how texts are loaded, presented, navigated and revisited, while the ORP remains the central visual feature of the reading display.

5 Evaluation of the RSVP Web Tool

5.1 User testing methodology

Usability testing should answer whether people can operate the tool and recover from problems. It is distinct from the experiment on reading performance. A user may find the interface easy to use while comprehending less at an excessive rate, or may understand the passage while struggling to locate a control. Both outcomes matter, but they support different conclusions.

A proposed formative study would recruit approximately six to eight adult readers with varied familiarity with RSVP. This is an initial practical round, not a statistically representative sample. Participants would complete realistic tasks: paste text, choose a rate, start, pause, resume, restart without automatic playback, seek to an earlier point, save a passage and reload it. Include tasks only for functions implemented in the tested version. Record independent success, success with help, failure, time and observed errors.

Think-aloud commentary can be used during interface tasks, but should be avoided during timed comprehension trials because speaking changes the task. After each task, ask a neutral ease-of-use question and follow with an open prompt about any difficulty. The interview should avoid leading questions such as asking whether ORP made reading easier. Instead, ask what participants noticed and what they would change.

Record the software version, device and testing context alongside anonymised observations. Separate direct participant statements from the researcher's interpretations. The study should include keyboard-only operation and at least one small-screen layout review. Such checks can identify practical problems, although they do not constitute comprehensive accessibility certification or establish suitability for all disabilities.

5.2 User feedback and results

No formal user-testing results or participant quotations were supplied. I used the annotations as development feedback and acted on them through the iterations described in Chapter 4. They are not findings from a recruited sample of users. The available record identifies restart behaviour, seeking, saved texts, a broken idle label, time information and spacing as concerns (Project design record, n.d.).

Some requested controls become visible in subsequent images. Direct speed entry and remaining-time information appear after being requested; the saved-text sidebar and Save controls appear after the sidebar proposal; and the latest image shows an intact Ready label. These are documentary observations about interface evolution. They do not establish successful task completion, satisfaction scores or a causal improvement in reading.

The participant results subsection remains to be completed with the actual sample and task outcomes. Report task success as both counts and denominators, since percentages alone can exaggerate precision in a small study. Present negative and ambiguous feedback as well as favourable comments. A proposed change should be traced to its supporting observation and checked again after implementation.

5.3 Evaluation of the tool

The tool can currently be assessed at the level of design rationale. The large reading area gives the presented word visual prominence. Progress and remaining-time indicators provide potential cues about status. Pause, Restart and seeking provide potential routes for control and recovery. These judgments apply Nielsen's (1994) heuristic principles to visible design elements; they are not empirical usability scores.

Three weaknesses remain especially important to test. The distinction between Start and Resume may be unclear unless their enabled states are consistent. A slider may allow movement through a passage while offering insufficient context to identify a desired sentence. Saved-text controls may encourage reliance on persistence that has not been demonstrated. Each issue should be converted into a task with an observable success criterion.

Overall effectiveness should be evaluated against multiple dimensions: comprehension, completion time, recovery, comfort and ease of operation. This multidimensional criterion is proposed by the thesis because the tool's intended value is practical reading. A high configured WPM or a polished screenshot cannot alone satisfy it. The available evidence supports the direction of the design, while leaving effectiveness open.

5.4 Limitations

The most substantial limitation is the absence of original participant data. This prevents claims about effect size, statistical significance, preference or usability success. The live website and source code were also inaccessible during preparation, preventing functional inspection and verification of technical implementation. These gaps should be resolved through an archived software version, recorded acceptance checks and actual study results.

The design record is selective and retrospective. Its selected images may omit rejected alternatives, intermediate changes and unresolved defects. Their numbering supplies an order but not development dates. The ChatGPT sidebar reference and the resulting RSVP sidebar document a design adaptation rather than independent evidence of end-user needs. There is also no independent evidence about mobile behaviour, persistence, timing accuracy or accessibility.

The literature review is not systematic and includes sources accessed at different levels, including full articles, abstracts and a publisher's book overview. Appendix B identifies these limits. It cannot establish the absence of newer contrary evidence. Before final submission, the review should be updated through the university's scholarly databases and full texts should be consulted where this draft relies on abstract-level findings.

5.5 Potential real-world applications

The most defensible immediate role for this project is as an exploratory reading interface and a research demonstrator. It allows presentation settings to be made explicit and, after appropriate instrumentation, could support comparisons of timing, colour and control. In this role, the value is the ability to investigate behaviour rather than a promise of superior reading performance.

Short, linear, low-stakes passages are a plausible area for further testing. A user might choose RSVP when they want guided pacing and can readily pause or return to normal text. This is a proposed use case, not a demonstrated benefit. Dense academic arguments, code, equations and documents requiring frequent cross-reference should be assessed separately because they create different information-access requirements.

Educational use should be similarly specific. RSVP could help demonstrate the distinction between exposure speed and comprehension, or provide material for an interface-design exercise. Boo and Conklin's (2015) findings caution against treating rapid word presentation as automatically beneficial for second-language comprehension. Any future learning application would need outcomes such as delayed understanding and transfer, not simply whether a passage finished quickly.

Specialist accessibility applications remain outside this project's evidence. User-adjustable settings may broaden choice, but choice alone does not prove therapeutic or assistive benefit. A targeted application should begin with the relevant users' tasks and evaluate the chosen presentation method against available alternatives.

5.6 Future improvements and features

ORP colour and alignment controls

My principal planned addition is a setting that allows the ORP colour to be changed or turned off. The setting should distinguish removing the colour from removing the alignment. A reader who disables highlighting should retain the same word position unless they independently change that option. A small set of clearly labelled colour choices and a no-highlight mode would provide a simple starting point. Contrast should be checked against the chosen background (W3C, 2024).

This addition would serve two purposes: personalisation and experimental separation of interface components. The settings could support comparison of an anchored monochrome word with an anchored highlighted word. Any claim of benefit should follow testing; the default should not imply that red is universally optimal. Settings should be previewable while playback is paused, so a change does not cause missed words.

Navigation and contextual reading

Sentence-back and sentence-forward controls would complement the coarse progress slider. A contextual view could show the current sentence or paragraph while paused, with a clear option to return to conventional text. Bookmarks and a visible resume position could help with interruptions. These proposals address the project's recovery requirements. They may also increase total reading time, which should be included in evaluation rather than hidden.

Adjustable timing

Optional punctuation delays and longer exposures for selected tokens could be introduced with a transparent explanation of their effect on average speed. An initial countdown could help users prepare before the first word

appears. A manually controlled mode could support difficult passages. Adaptive timing based on comprehension checks is a longer-term possibility, but would require evidence that adaptation improves outcomes and does not simply make the system harder to predict.

Reading view and appearance

A dedicated reading view could reduce the visual prominence of the title and text-entry area once playback begins. Adjustable font size, light and dark themes, and controlled vertical placement would address concerns seen in the final images. The user should retain an obvious route back to editing. This proposal responds to the positioning problem more directly than relying exclusively on extra page padding.

Text management and research support

Saved passages could support titles, search, export, deletion and explicit recovery after a failed save. A privacy description should match the actual implementation. Research functionality could add consent-based event logging, software version identifiers, condition settings and timestamp export. These features would make the tool easier to evaluate, but participant data collection should remain separate from ordinary use and should not be added invisibly.

6 Discussion and Conclusion

6.1 Overall findings

This thesis brings together published research, a documented design process and a proposed evaluation. Its strongest completed project finding is that the supplied development record repeatedly moves towards greater reader control: pausing at restart, revisiting text, estimating remaining time and returning to saved passages. Later images show attention shifting towards visual hierarchy and comfortable placement. These findings concern the design process, not participant performance.

The literature provides a reason to judge RSVP by more than presentation speed. Different studies used different readers, materials and tasks, and reported outcomes that cannot be reduced to a universal optimum. The project therefore adopts comprehension and recoverability as explicit criteria alongside completion time. This is a design conclusion informed by the reviewed evidence, rather than proof that the present interface already meets those criteria.

The experimental proposal makes the outstanding uncertainties measurable. It separates rate from format where practicable, distinguishes self-reported focus from comprehension, and treats ORP colour and alignment as independent variables. The numerical differences in Chapter 3 are calculated from explicit assumptions and are not original empirical effect estimates. The absence of a dataset limits the conclusion, but it does not prevent a clear account of what needs testing and why.

6.2 The future of RSVP

The future development proposed here is centred on choice and integration. A reader should be able to use a serial stream when it suits the task, switch to contextual text when needed and recover from interruptions without losing position. This is a prospective design direction, not a prediction of commercial adoption. It gives the project a practical purpose even if it does not outperform conventional reading in every comparison.

Personalisation also needs a clear evidential basis. A colour setting can accommodate preference without being described as a cognitive enhancement. A timing adaptation can be tested against a transparent fixed-rate baseline. A longer-term study could examine whether users return voluntarily, which passages they choose and whether comprehension changes with practice. Recording task choice would help explain usefulness more directly than asking participants to imagine future use.

6.3 Answering the research questions

What are the effects of RSVP on reading, comprehension and attention? The reviewed evidence shows that RSVP changes access to text by controlling the sequence and timing of presentation. Reading-rate gains have been observed in some experimental settings, while comprehension costs have been reported in others, particularly under rapid presentation. The specific evidence and limits are set out in Chapter 2. No effect of this project's tool on comprehension or attention has yet been measured. A feeling of focus should therefore be reported as an experience rather than treated as demonstrated understanding.

Where and how can RSVP be effectively implemented? Its established research role is supported by controlled studies of rapid visual processing. For the present web tool, plausible applications require a defined reading task, manageable timing, dependable pause and recovery controls, and access to the original text. The proposed use cases remain to be evaluated. The most useful implementation is not necessarily the one with the highest selectable WPM, but the one whose outcomes meet the reader's purpose under tested conditions.

The answer to both questions is consequently conditional. Effectiveness depends on what information must be retained, how much control the reader has and how performance is measured. The proposed study provides a route to replacing those conditions with project-specific evidence.

6.4 Conclusion

The project began with curiosity about an online reading-speed test and developed into an investigation of the relationship between presentation, understanding and control. Building a basic RSVP reader and adding ORP created a practical context for these questions. The supplied images document a process of identifying problems and refining the interface through navigation, feedback, text management and layout decisions.

The resulting thesis draft supports a measured design position: RSVP deserves evaluation as an optional reading method, and its success should be demonstrated through comprehension, usable control and appropriate timing.

ORP colour customisation is a worthwhile next addition because it supports choice and permits a cleaner experimental comparison. The essential next research step is to complete the proposed studies and report their actual outcomes. Until then, the web tool is a documented design project with a clear evaluation plan, rather than a proven improvement to reading performance and can be utilised in any further independent research.

For a project concerned with reading faster, the final lesson may be that evaluation itself should not be rushed.

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